

Thermodynamics

Lecture 3A:

- Pure substances
- Property tables

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After this lecture

- Introduction to pure substances and phase change processes.
- Illustrate (p,v) , (T,v) and (p,T) diagrams and (p,v,T) surfaces of pure substances.
- Demonstrate the procedures for determining thermodynamic properties of pure substances.

Learning objectives

What you will be able to do after this course:

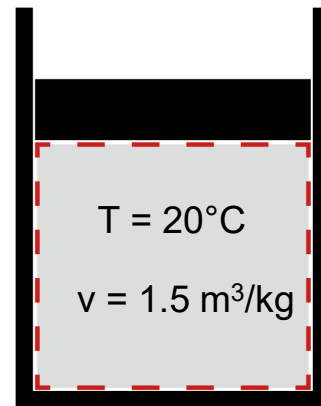
- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. *Comprehension:* Describe the first and second laws in the student's own language.**
- III. *Application:* Solve simple single-answer problems using the first law.**
- IV. *Analysis:* Solve problems requiring both the first and second laws.
- V. *Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

Learning objectives (II)

- VI. Evaluation: Determine and describe second law fallacies in proposed power cycles.
- VII. Evaluation: Judge when classical thermodynamics is not the appropriate analysis tool.
- VIII. Evaluation: Find and correct errors in the student's own solutions and in those of others.
- IX. *Analysis and Evaluation*: Search appropriate databases and the literature to find required thermodynamic data, and if the data are not available the student can select appropriate procedures and predict the values of the data.
- X. Knowledge: Briefly outline the history of the field of thermodynamics.

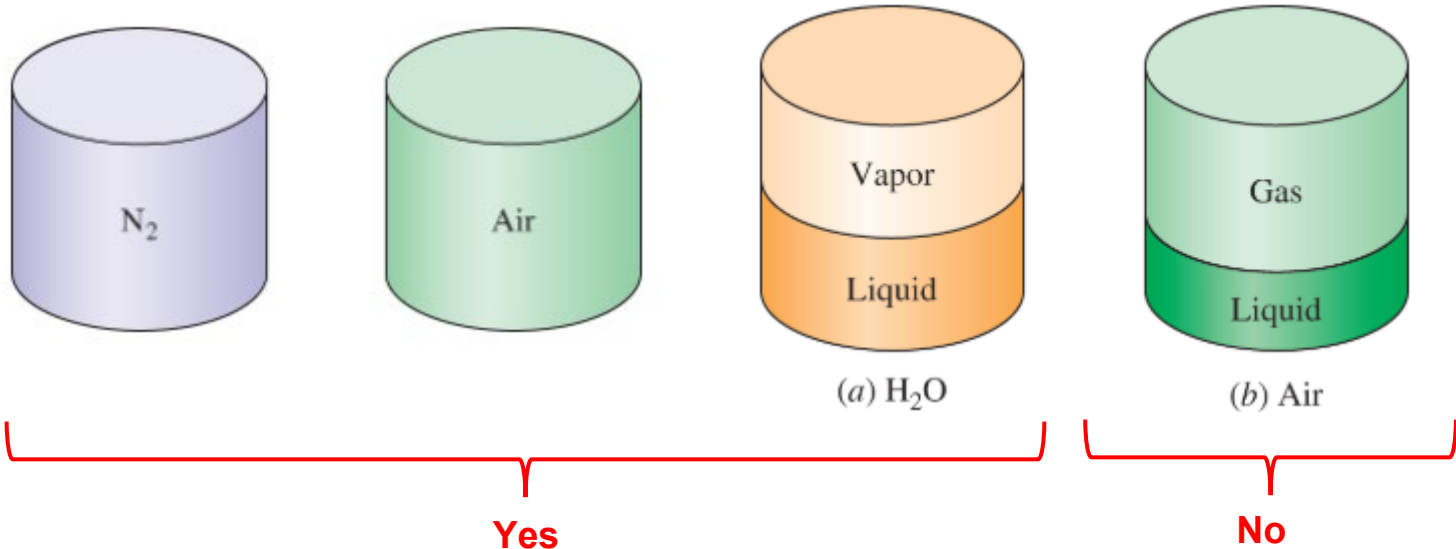
Properties of pure substances

- Thermodynamic description of the working fluids:
 - Water
 - Refrigerants (Ammonia, R134a, ...)
 - Gases – (often approximated as ideal gases)
 - Hydrocarbons
- Studying thermodynamic properties of different phases of matter.



Pure substances

- Definition: a fluid with a uniform chemical composition



Phases of matter

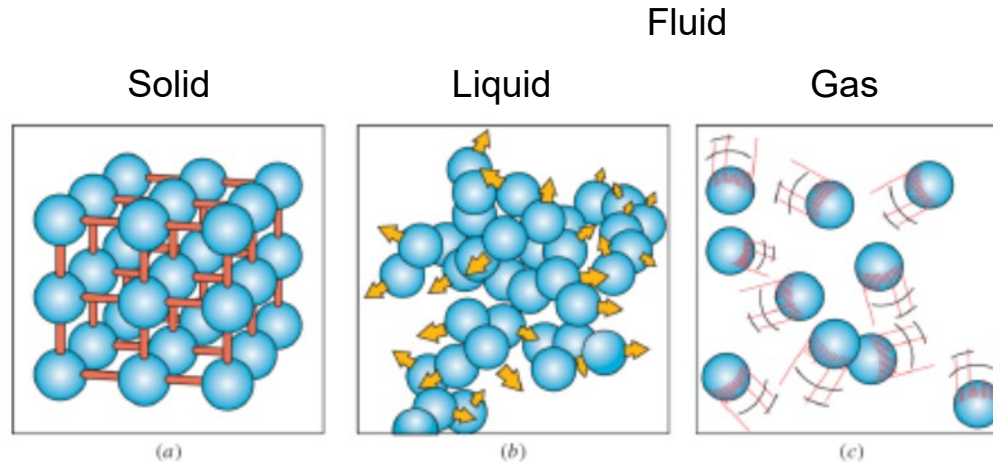


FIGURE 3–4

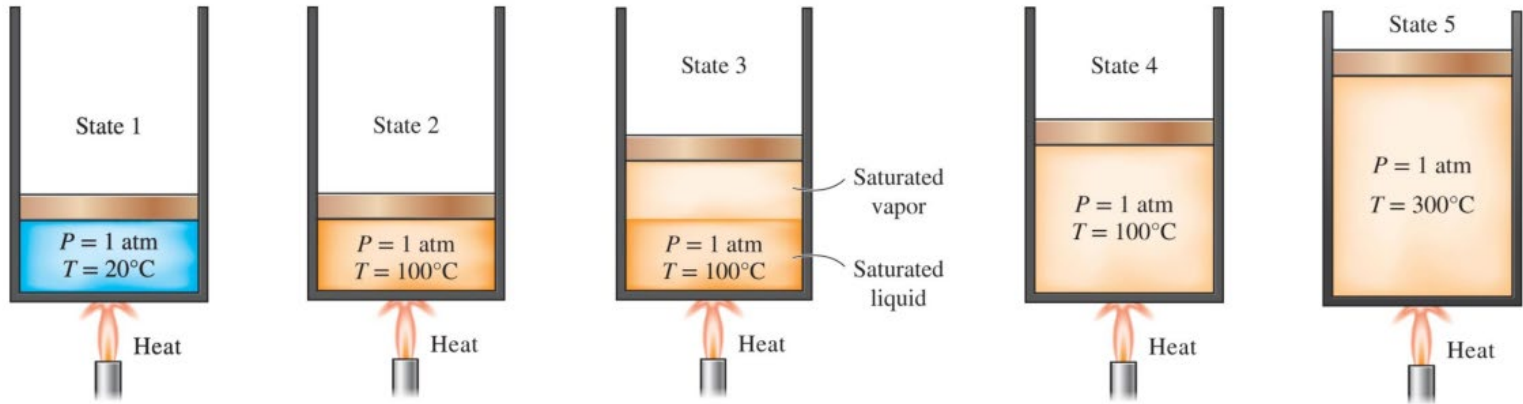
Melting
Freezing

Vaporizing
Condensation

Sublimation
Deposition (resublimation)

Phase change processes of pure substances

- Example: isobaric heating of Water



Compressed liquid
Subcooled liquid

Saturated liquid

Saturated liquid-
Vapor mixture

Saturated vapor

Superheated vapor

Not about to vaporize

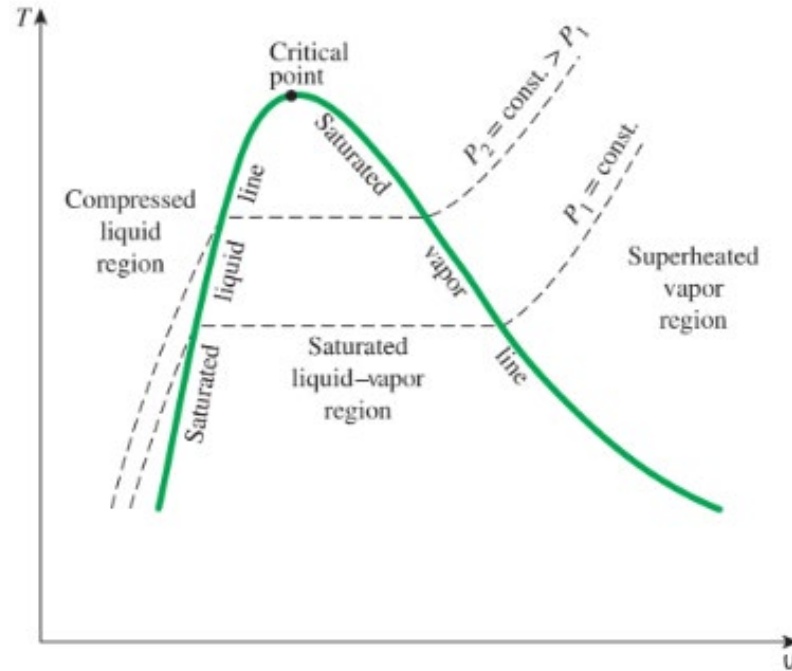
About to vaporize

Phases co-exist

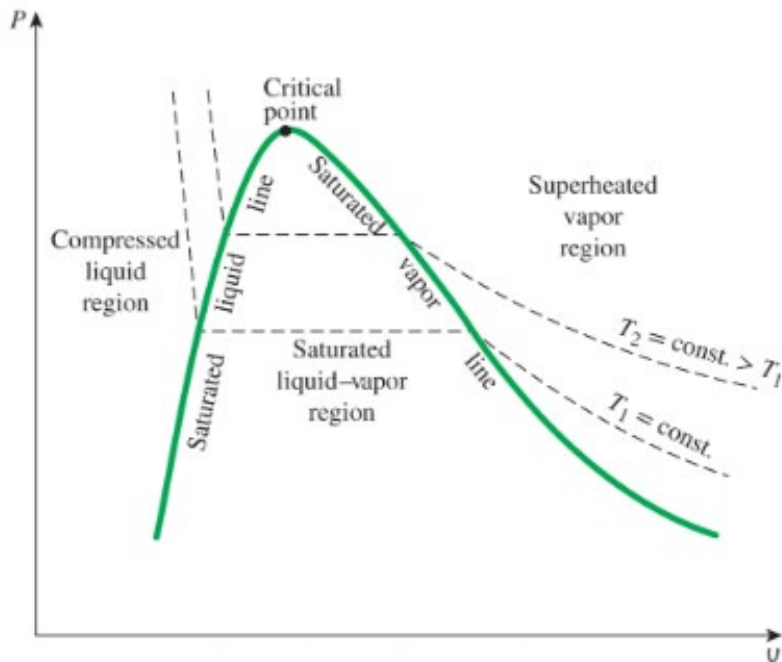
About to condense

Not about to condense

T-v diagram for pure substances



P-v diagram for pure substances

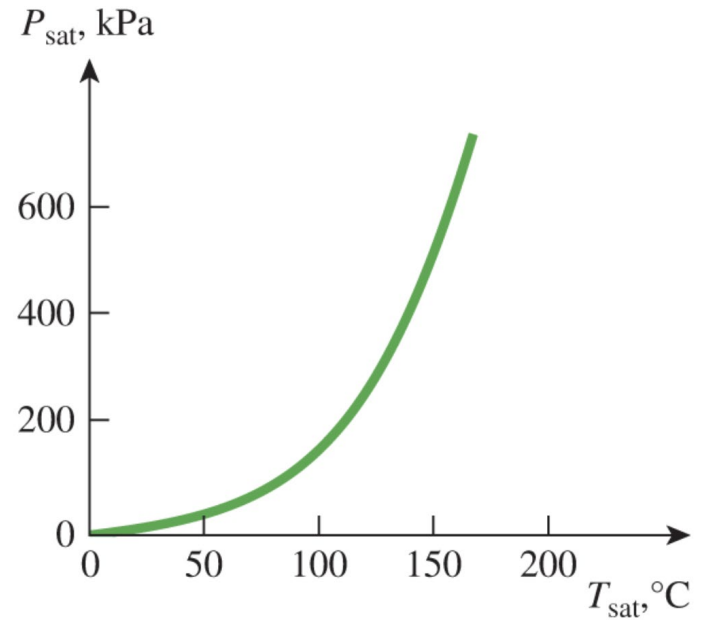


P-v diagram of a pure substance

Saturation temperature and pressure

- P_{sat} pressure at which a pure substance changes phase at a given temperature.
- T_{sat} temperature at which a pure substance changes phase at a given pressure.
- During phase change process Pressure and temperature are linked.

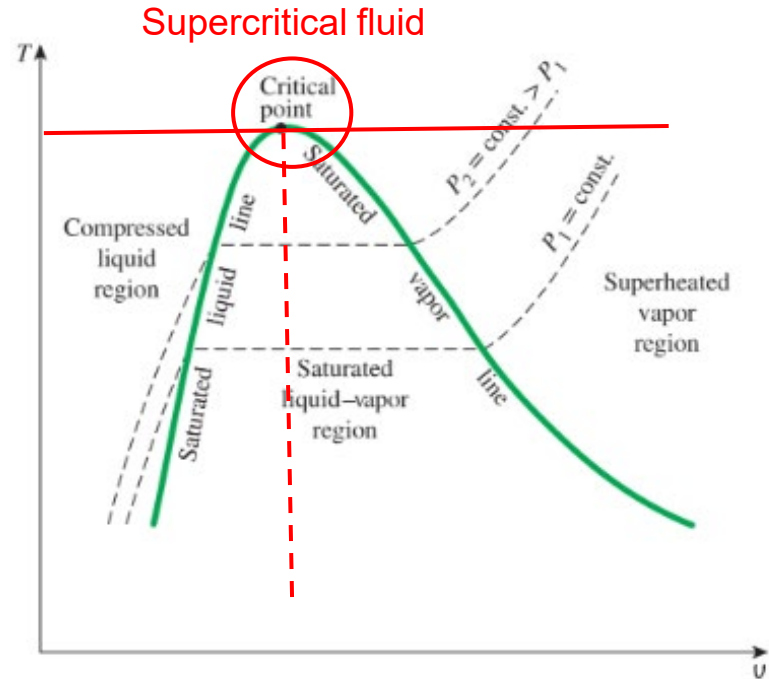
$$P_{sat} = f(T_{sat})$$



Liquid vapor saturation curve

Critical point

- At pressures above critical point: only one phase ever, resembles vapor, but cannot tell when phase change occurred.



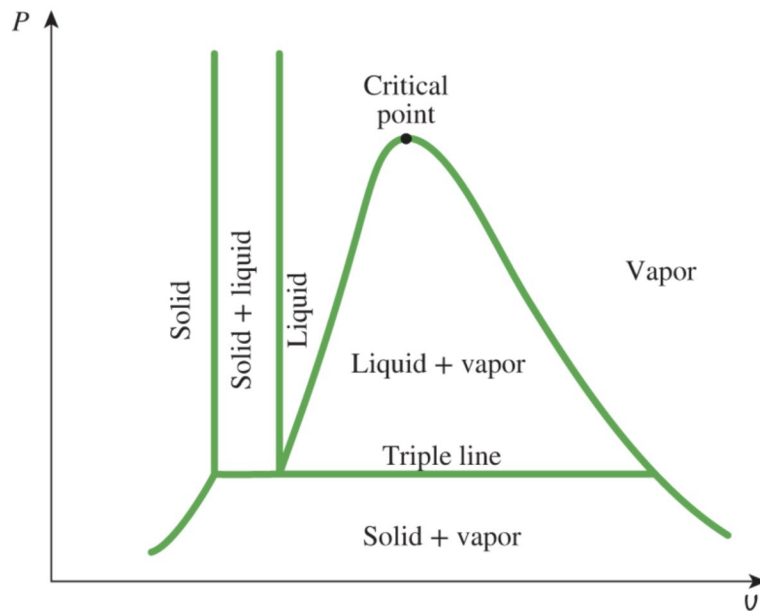
T-v diagram of a pure substance

Demonstration critical point of CO₂



Solid-Liquid-Vapor Diagram

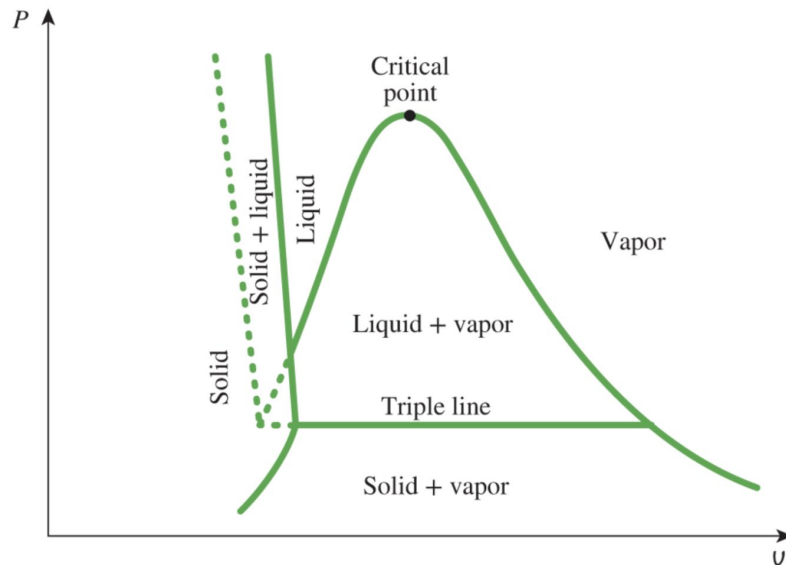
- Triple line: solid, liquid and vapor phases coexist.
- $T_{trip} = 273.16K, p_{trip} = 0.61kPa$
- Fluids can either contract or expand upon freezing.



P-v diagram of a pure substance that contracts upon freezing

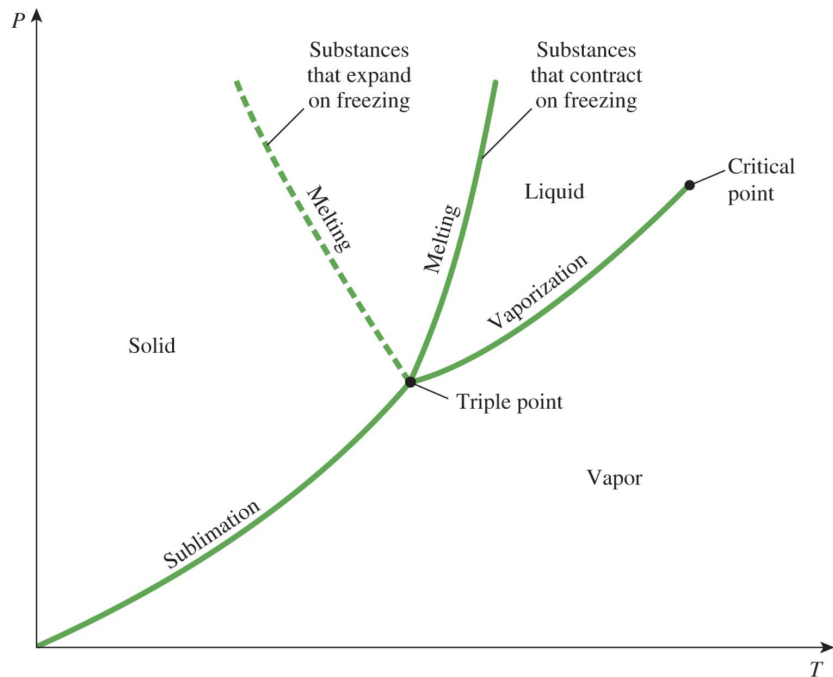
Solid-Liquid-Vapor Diagram

- Few substances expand upon freezing.
- For example, water.



P-v diagram of a pure substance that expands upon freezing

P-T diagram for pure substances

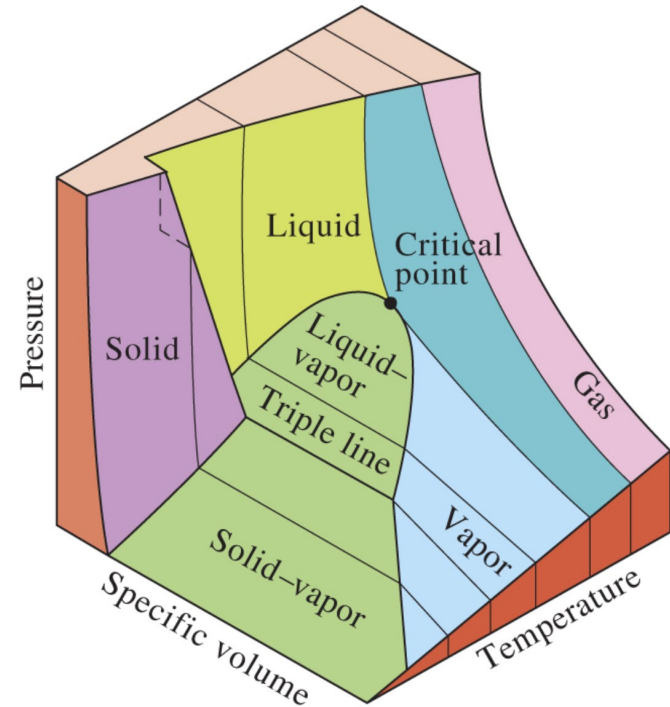


P-T diagram of a pure substance

The P-T-v surfaces in 3D

- State postulate: state of a simple compressible substance is fixed by any two independent intensive properties.

$$P = f(T, v)$$



Thermodynamic property tables

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Thermodynamic property tables

TABLE A-4

Saturated water—Temperature table

Temp., T °C	Sat. Press., P_{sat} kPa	Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, $\text{kJ/kg}\cdot\text{K}$		
		Sat. liquid, u_f	Sat. vapor, u_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
0.01	0.6117	0.001000	206.00	0.000	2374.9	2374.9	0.001	2500.9	2500.9	0.0000	9.1556	9.1556
5	0.8725	0.001000	147.03	21.019	2360.8	2381.8	21.020	2489.1	2510.1	0.0763	8.9487	9.0249
10	1.2281	0.001000	106.32	42.020	2346.6	2388.7	42.022	2477.2	2519.2	0.1511	8.7488	8.8999
15	1.7057	0.001001	77.885	62.980	2332.5	2395.5	62.982	2465.4	2528.3	0.2245	8.5559	8.7803
20	2.3392	0.001002	57.762	83.913	2318.4	2402.3	83.915	2453.5	2537.4	0.2965	8.3696	8.6661
25	3.1698	0.001003	43.340	104.83	2304.3	2409.1	104.83	2441.7	2546.5	0.3672	8.1895	8.5567
30	4.2469	0.001004	32.879	125.73	2290.2	2415.9	125.74	2429.8	2555.6	0.4368	8.0152	8.4520
35	5.6291	0.001006	25.205	146.63	2276.0	2422.7	146.64	2417.9	2564.6	0.5051	7.8466	8.3517
40	7.3851	0.001008	19.515	167.53	2261.9	2429.4	167.53	2406.0	2573.5	0.5724	7.6832	8.2556
45	9.5953	0.001010	15.251	188.43	2247.7	2436.1	188.44	2394.0	2582.4	0.6386	7.5247	8.1633
50	12.352	0.001012	12.026	209.33	2233.4	2442.7	209.34	2382.0	2591.3	0.7038	7.3710	8.0748
55	15.763	0.001015	9.5639	230.24	2219.1	2449.3	230.26	2369.8	2600.1	0.7680	7.2218	7.9898

Property tables: Saturated liquid and vapor

Temp., T °C	Sat. Press., P_{sat} kPa	Specific volume, m^3/kg	
		Sat. liquid, v_f	Sat. vapor, v_g
0.01	0.6117	0.001000	206.00
5	0.8725	0.001000	147.03
10	1.2281	0.001000	106.32
15	1.7057	0.001001	77.885
20	2.3392	0.001002	57.762

Subscript f: properties of saturated liquid

Subscript g: properties of saturated vapor

v_f specific volume of liquid (m^3/kg)

v_g specific volume of vapor (m^3/kg)

Property tables: Saturated liquid and vapor

Temp., T °C	Sat. Press., P_{sat} kPa	Internal energy, kJ/kg		
		Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g
0.01	0.6117	0.000	2374.9	2374.9
5	0.8725	21.019	2360.8	2381.8
10	1.2281	42.020	2346.6	2388.7
15	1.7057	62.980	2332.5	2395.5
20	2.3392	83.913	2318.4	2402.3

Subscript f: properties of saturated liquid

Subscript g: properties of saturated vapor

u_f specific internal energy of liquid (kJ/kg)

u_g specific internal energy of vapor (kJ/kg)

u_{fg} specific internal energy of vaporization

Property tables: Saturated liquid vapor mixture

Given: m_f, m_g determine v_{avg}

$$V_{total} = m_f v_f + m_g v_g$$

$$m_f = m_{total} - m_g$$

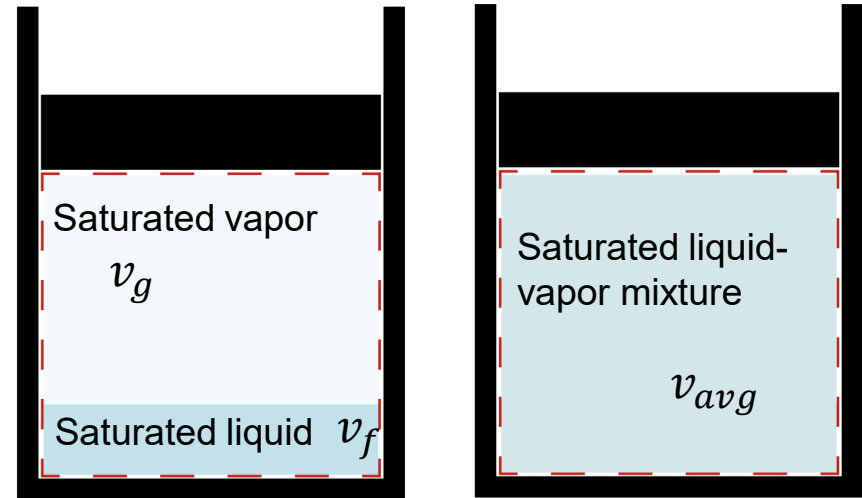
$$V_{total} = (m_{total} - m_g) v_f + m_g v_g$$

$$V_{total} = m_t v_{avg}$$

$$\text{Vapor quality } x \quad x = \frac{m_g}{m_{total}}$$

$$v_{avg} = (1 - x) v_f + x v_g$$

$$v_{avg} = v_f + x v_{fg}$$



Summary

- Introduction to pure substances and phase change processes.
- Illustrate (p,v) , (T,v) and (p,T) diagrams and (p,v,T) surfaces of pure substances.
- Demonstrate the procedures for determining thermodynamic properties of pure substances.

Next up

- Description of ideal gasses.